

2006

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 19, 23, 25 and 27

Total marks – 100

Section I Pages 2–28

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 29–44

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

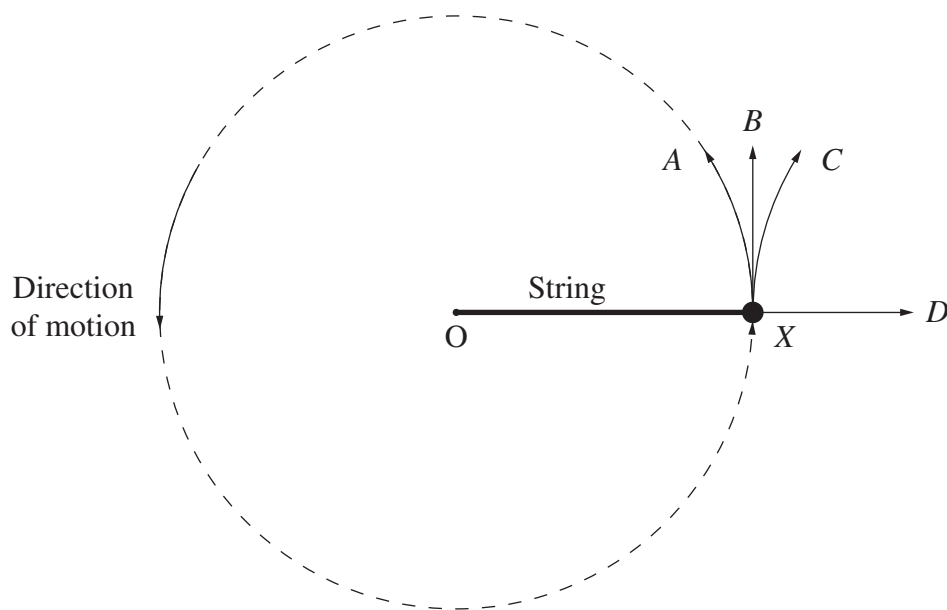
If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct →

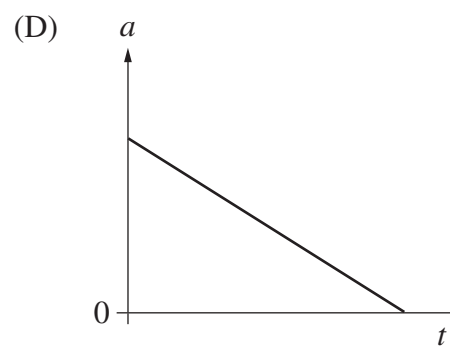
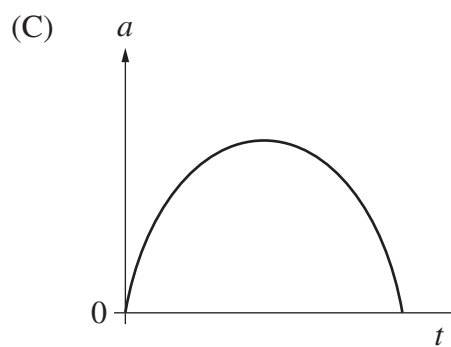
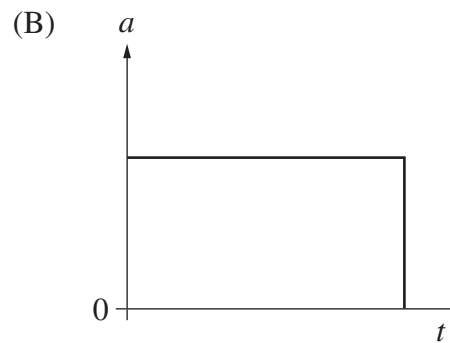
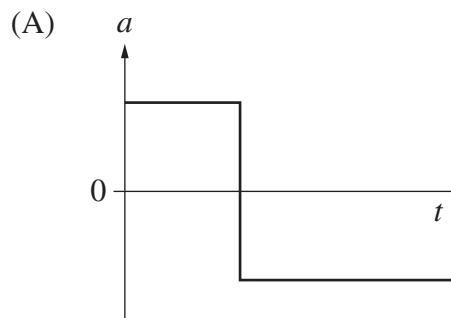
- 1 Given that G is the universal gravitational constant, and g is the magnitude of the acceleration due to gravity, which statement is true?
- (A) The values of G and g depend on location.
- (B) The values of G and g are independent of location.
- (C) G is the same everywhere in the universe, but g is not.
- (D) g is the same everywhere in the universe, but G is not.
- 2 A mass attached to a length of string is moving in a circular path around a central point, O , on a flat, horizontal, frictionless table. This is depicted in the diagram below. The string breaks as the mass passes point X .



- Which line best depicts the subsequent path of the mass?
- (A) Line A
- (B) Line B
- (C) Line C
- (D) Line D
- 3 What is the main reason why the Michelson-Morley experiment is considered important?
- (A) It shows the existence of the aether.
- (B) It suggests that light is an electromagnetic wave.
- (C) It indicates that light can exhibit interference effects.
- (D) It provides experimental support for the theory of relativity.

- 4 A stone is thrown horizontally from the top of a cliff and falls onto the beach below.

Which acceleration–time graph best describes the motion of the stone?

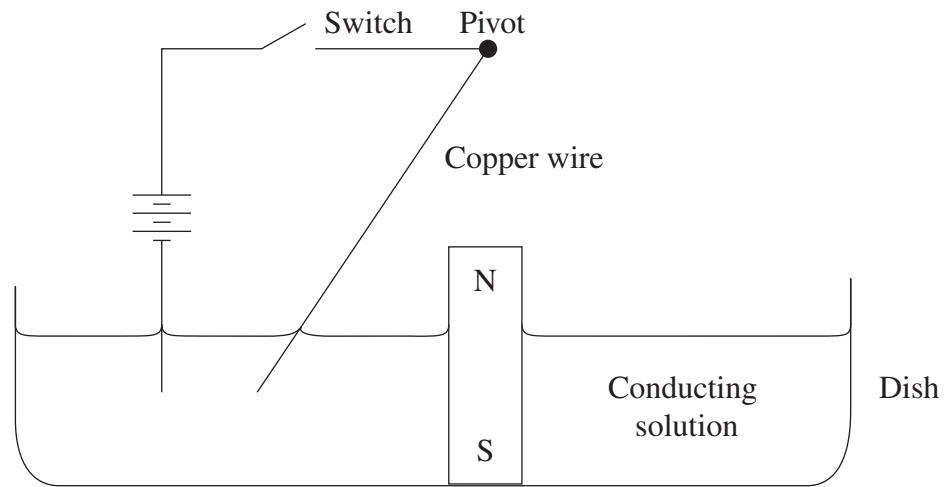


- 5 Two satellites, X and Y , are in circular orbits around Earth. Their masses are identical and their orbital radii are R and $16R$, respectively.

What is the ratio of their orbital periods, $T_X : T_Y$?

- (A) 1 : 4
(B) 1 : 16
(C) 1 : 32
(D) 1 : 64

- 6 The diagram shows a magnet standing on the bottom of a dish filled with a conducting solution. A copper wire is suspended freely from a point above the magnet with its tip in the conducting solution. It is held in the position shown.

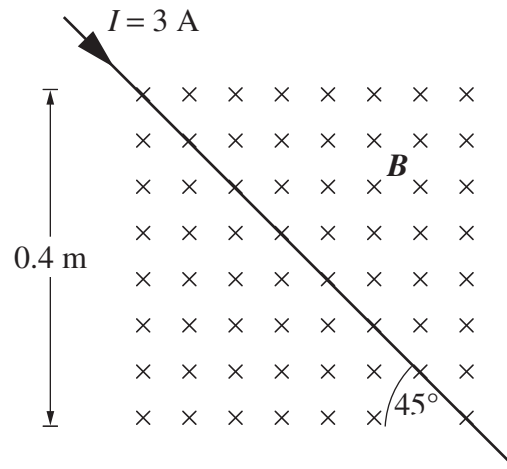


The switch is closed and the wire released.

Which of the following will be observed?

- (A) The wire will rotate about the magnet.
- (B) The wire will be attracted to the magnet.
- (C) The magnet will rotate about its vertical axis.
- (D) The solution in the dish will rotate about the magnet.

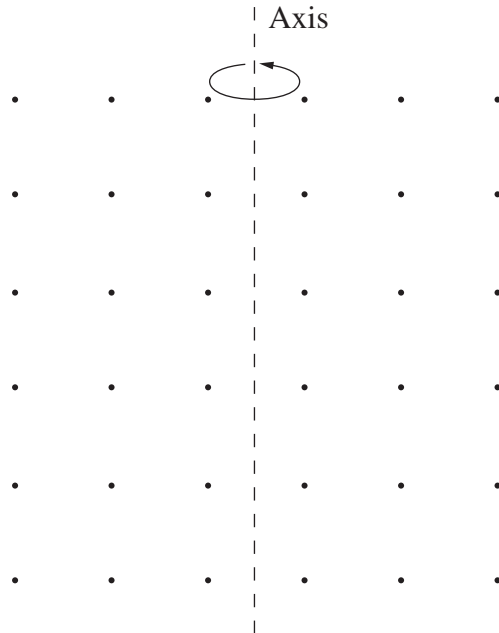
- 7 A current-carrying conductor passes through a square region of magnetic field, magnitude 0.5 T, as shown in the diagram. The magnetic field is directed into the page.



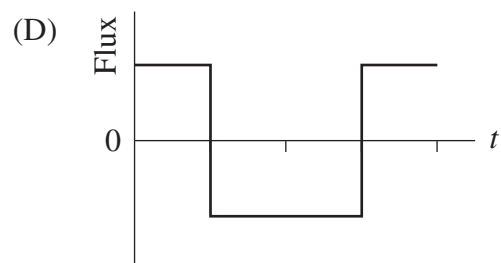
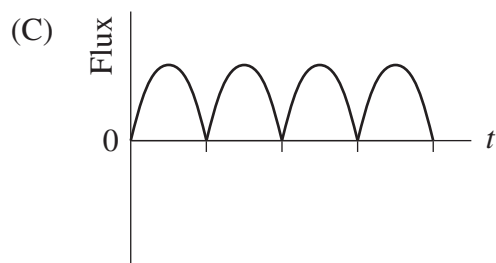
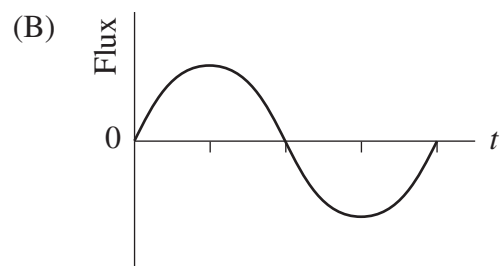
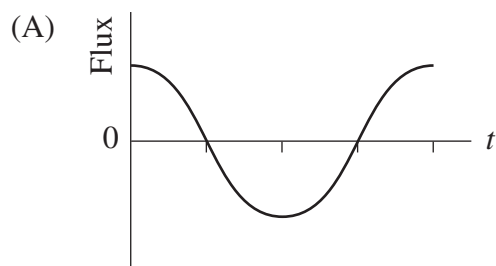
What is the magnitude of the magnetic force on the conductor?

- (A) 0.170 N
- (B) 0.424 N
- (C) 0.600 N
- (D) 0.849 N

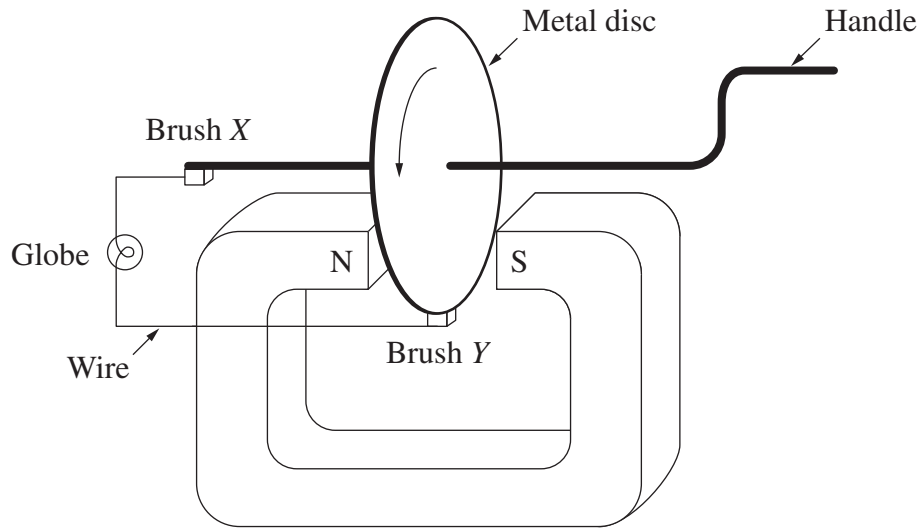
- 8 A square loop of wire, in a uniform magnetic field, is rotating at a constant rate about an axis as shown. The magnetic field is directed out of the plane of the page. At time $t=0$ the plane of the loop is perpendicular to the magnetic field and side XY is moving out of the page.



Which graph best represents the variation of the magnetic flux through the loop with time?



- 9 Early electric generators were often very simple. A hand-operated version is depicted below.

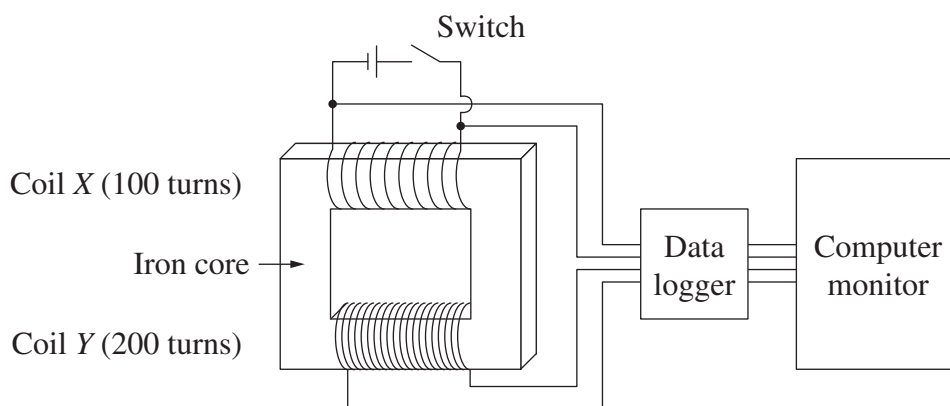


Brush *X* touches the metal axle and Brush *Y* touches the rim of the disc.

If the metal disc is rotated uniformly as shown, which statement about the current through the globe is correct?

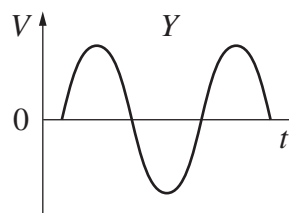
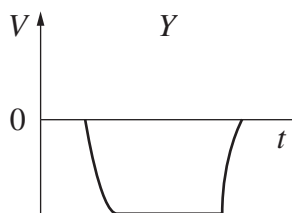
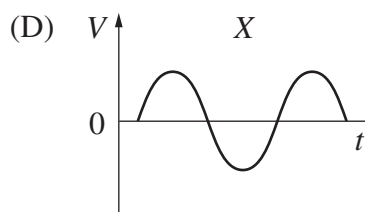
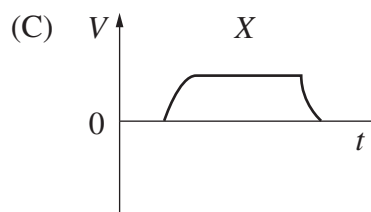
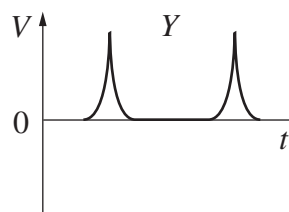
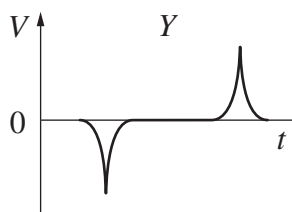
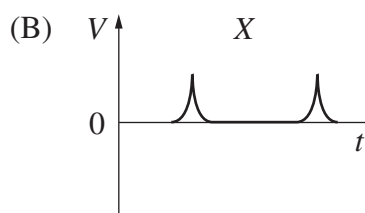
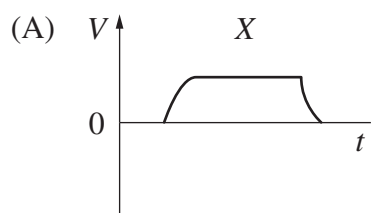
- (A) No current flows.
- (B) A direct current flows from *Y* to *X*.
- (C) A direct current flows from *X* to *Y*.
- (D) An alternating current flows between *X* and *Y*.

- 10 The apparatus shown is designed to investigate the operation of a transformer.



A student closes the switch for a short time, then opens it. The data logger records values of voltage for both coils for the duration of the investigation. The data logger software displays the results as a pair of voltage–time graphs on a computer monitor.

Which pair of graphs best depicts the student's results?



- 11** Lawrence and William Bragg used X-rays to determine the crystal structure of materials.

Which property of waves was the basis of their technique?

- (A) Diffraction
- (B) Dispersion
- (C) Polarisation
- (D) Rarefaction

- 12** A charged non-magnetic particle is moving in a magnetic field.

What would NOT affect the magnetic force on the particle?

- (A) The strength of the magnetic field
- (B) The magnitude of the charge on the particle
- (C) The velocity component parallel to the magnetic field direction
- (D) The velocity component perpendicular to the magnetic field direction

- 13** The temperature of a metal is reduced.

Which statement correctly identifies the change in its electrical resistance and the reason for this change?

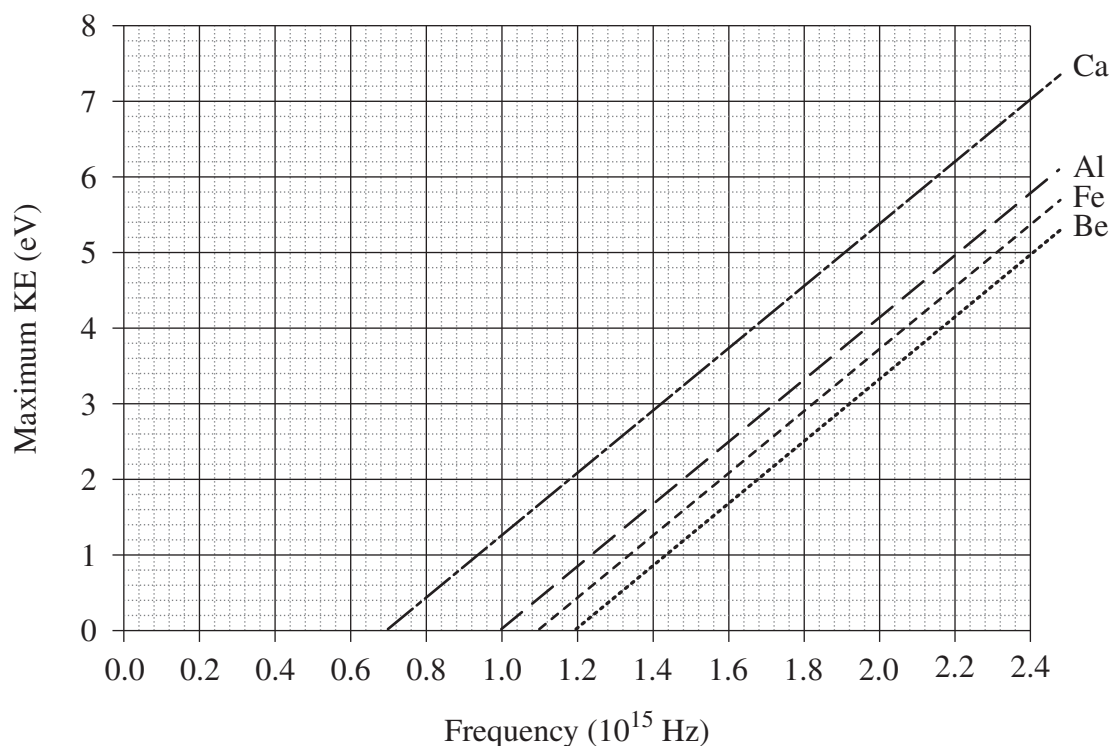
	<i>Electrical resistance</i>	<i>Reason</i>
(A)	Increases	Metal freezes
(B)	Decreases	More free electrons available
(C)	Increases	Electrons move more slowly
(D)	Decreases	Reduced metal lattice vibrations

- 14** A potential difference of 50 V is applied between two identical, parallel aluminium plates which are separated by a distance of 10 mm.

In order to double this electric field strength, which new arrangement should be used?

	<i>Separation (mm)</i>	<i>Potential difference (V)</i>	<i>Plates</i>
(A)	20	100	Aluminium
(B)	5	50	Perspex
(C)	10	100	Copper
(D)	20	50	Aluminium

- 15 When electromagnetic radiation shines on metals, photoelectrons may be emitted. The maximum kinetic energy of emitted photoelectrons is plotted against radiation frequency for four metals as shown in the graph.



Electromagnetic radiation of wavelength 187 nm shines upon an unknown metal and the maximum kinetic energy of the photoelectrons is found to be 2.5 eV.

Based on this information, what is the unknown metal?

- (A) Al
- (B) Be
- (C) Ca
- (D) Fe

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Centre Number

Section I (continued)

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Part B – 60 marks

Student Number

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

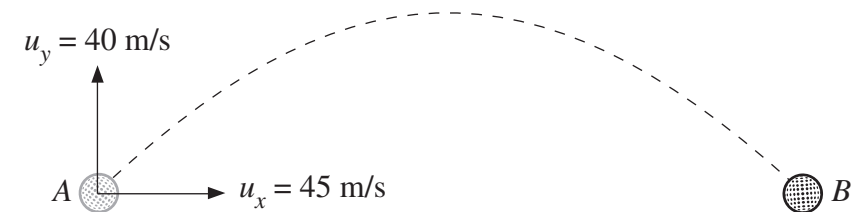
Show all relevant working in questions involving calculations.

Question 16 (6 marks)

Please turn over

Question 16 (6 marks)

A projectile leaves the ground at point A with velocity components as shown in the diagram. It follows the path given by the dotted line and lands at point B .



- (a) State the horizontal component of the projectile's velocity when it lands. 1

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- (b) Find the magnitude of the initial velocity of the projectile. 1

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- (c) Calculate the maximum height attained by the projectile. 2

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- (d) Calculate the range of the projectile, if it lands level with its starting position. 2

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Question 17 (6 marks)

Parts of a space mission involve a spacecraft spending time in geostationary orbit, and then returning safely to Earth.

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Analyse the forces acting on this spacecraft during these parts of the mission.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 18** (3 marks)

An object is stationary in space and located at a distance 10 000 km from the centre of a certain planet. It is found that 1.0 MJ of work needs to be done to move the object to a stationary point 20 000 km from the centre of the planet.

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Calculate how much more work needs to be done to move the object to a stationary point 80 000 km from the centre of the planet.

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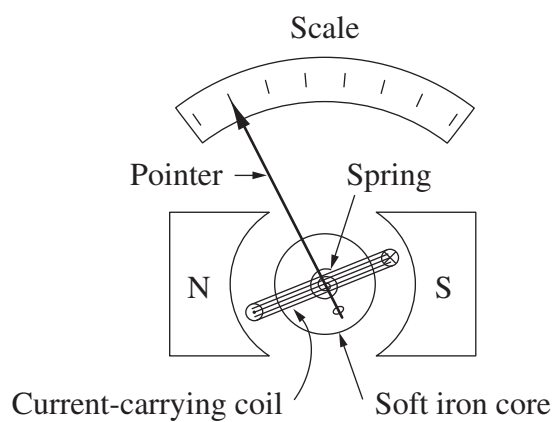
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Question 19 (3 marks)

The diagram shows the structure of a typical galvanometer.

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Describe how the galvanometer operates as an application of the motor effect.

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Centre Number

Section I – Part B (continued)

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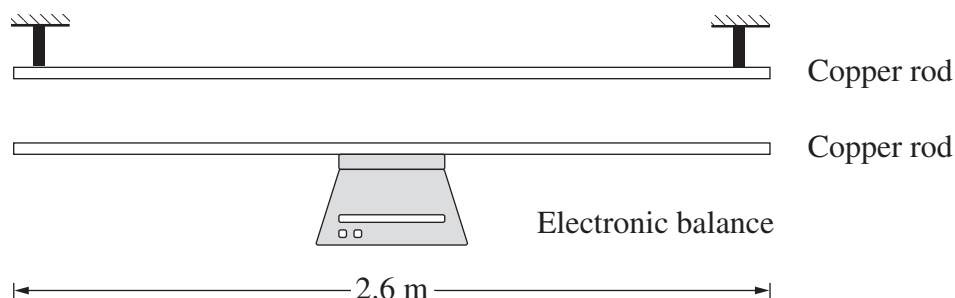
Student Number

Question 20 (8 marks)

Please turn over

Question 20 (8 marks)

A balance was used to investigate the relationship between current and force. The balance was set up with one copper rod fixed to it and a second rod fixed above it, as shown in the diagram. Each rod was connected to a source of current. The diagram is not to scale.



The copper rods were rigid, each was 2.6 m long, and they were parallel. The current in the upper rod was kept constant at 50 A. Different currents were passed through the lower rod and the balance reading recorded for each current. The readings are given in the table below.

<i>Current in lower rod</i> (A)	<i>Balance reading</i> (kg)
2.8	0.5485
8.0	0.5480
12.2	0.5474
16.8	0.5470
20.0	0.5465

- (a) Identify the relative directions of the currents in both rods, and justify your answer.

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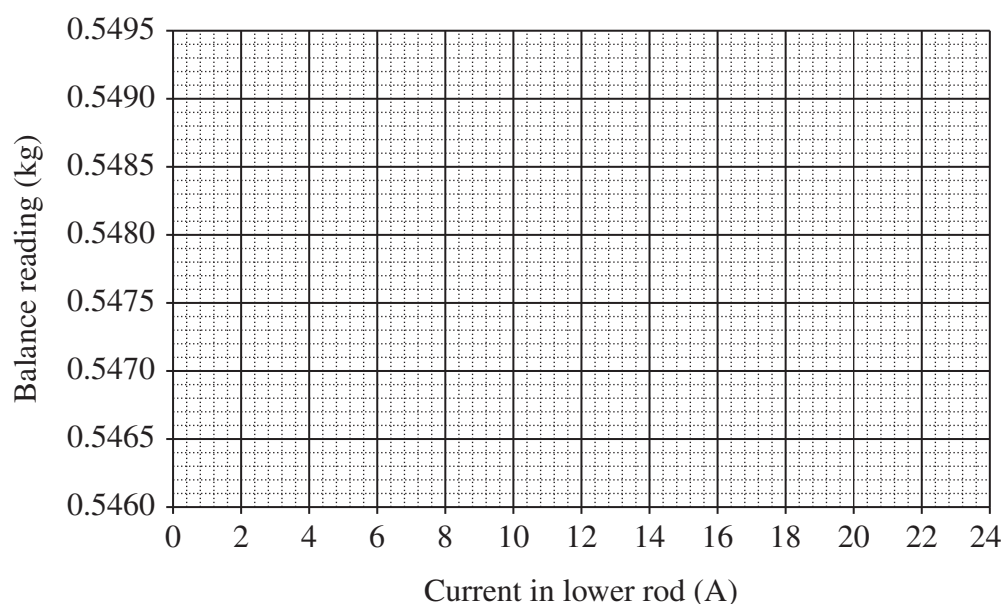
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Question 20 continues on page 21

Question 20 (continued)

- (b) Plot the data from the table onto the graph, using the scales and axes as indicated, and add the line of best fit (trend line). 2



- (c) Find the mass of the copper rod on the balance. 1

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- (d) Calculate the distance between the two copper rods. 3

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End of Question 20

Question 21 (6 marks)

Assess the impact on society and the environment of the potential applications of superconductors.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 22** (5 marks)

A student drops a bar magnet onto a large block of copper resting on the floor. The magnet falls towards the copper, slowing down as it comes close, then landing gently.

- (a) Explain the physics responsible for this observation.

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- (b) Predict what will happen if the experiment is repeated with a copper block cooled to approximately -50°C . Justify your prediction.

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Question 23 (6 marks)

- (a) Draw labelled diagrams of the band structures of an insulator, a semiconductor, and a conductor. 2

- (b) With reference to your diagrams, describe the differences in electrical resistance between insulators, semiconductors and conductors. 2

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- (c) Explain how the addition of trace amounts of certain elements, such as phosphorus, can change the electrical resistance of semiconductors at a given temperature. 2

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 24 (3 marks)

Discuss the origins of unwanted heat production in transformers and ways in which these can be overcome.

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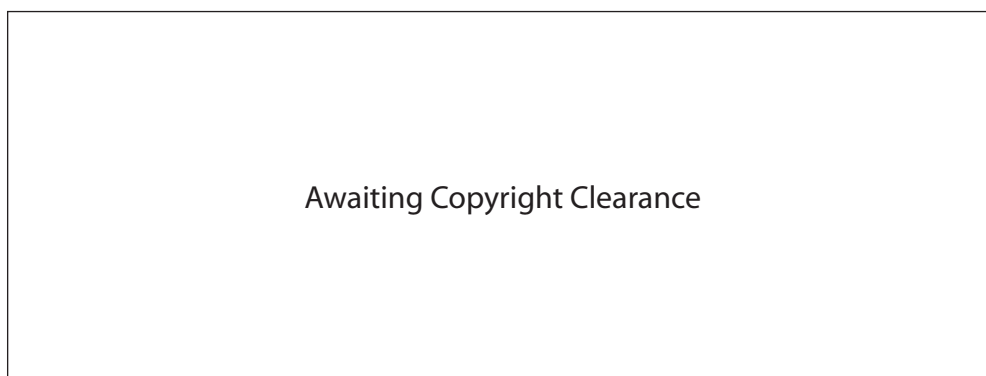
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Please turn over

Question 25 (6 marks)

A simplified cathode ray oscilloscope is depicted below.



- (a) Outline the roles of the deflection plates and the electrodes in the electron gun. 2

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- (b) In a special investigation, the voltage between the cathode and the anode is increased so that an electron gains a velocity of $0.60c$, where c is the speed of light. The electron starts from rest at the cathode. 2

Calculate the mass of this electron in the laboratory frame of reference.

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- (c) The distance between the anode and the screen, as measured in the electron's frame of reference, is 0.24 m . 2

Calculate this distance as measured in the laboratory frame of reference.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 26** (4 marks)

Beginning in the late 19th century, observations and experiments on black body radiation and the photoelectric effect led physicists to revise their existing model of light.

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Use the above as an example to explain how scientists test and validate models.

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Question 27 (4 marks)

J. Plücker was the first to observe cathode rays within gas discharge tubes. He inferred that the rays were a form of electromagnetic radiation.

- (a) Describe ONE subsequent observation that led other scientists to argue that cathode rays were charged particles. 2

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- (b) Identify ONE potential hazard associated with performing discharge tube investigations, and outline ONE safe work practice which addresses this hazard. 2

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Physics

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

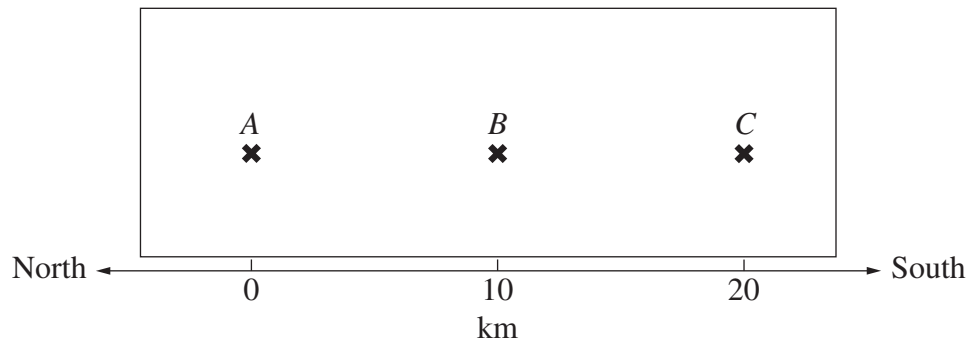
	Pages
Question 28 Geophysics	30–34
Question 29 Medical Physics	35–36
Question 30 Astrophysics	37–38
Question 31 From Quanta to Quarks	39–40
Question 32 The Age of Silicon	41–44

Question 28 — Geophysics (25 marks)

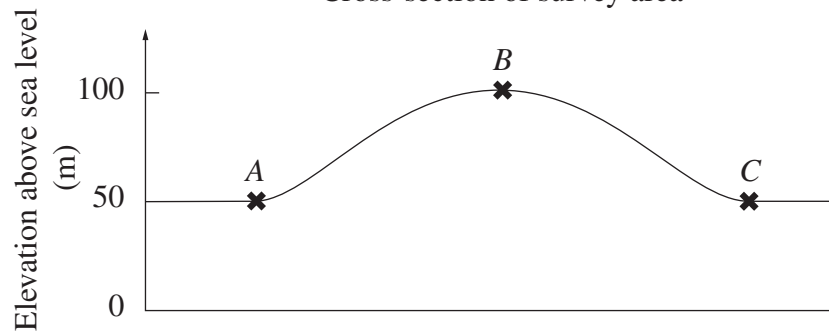
- (a) Gravimetric surveys collect raw gravity data which must be further processed by the application of a number of corrections. The following data table, map, and cross-section are the result of a small-scale gravimetric survey.

<i>Station</i>	<i>Gravitational reading (raw data) (mgal)</i>
<i>A</i> (datum station)	979158.1
<i>B</i>	979161.6
<i>C</i>	979174.2

Map of survey area



Cross-section of survey area



Question 28 continues on page 31

Question 28 (continued)

Gravity data reduction involves the application of the following corrections:

<i>Quantity</i>	<i>Correction</i>	<i>Application comment</i>
Latitude correction	0.8 mgal/km	Subtract a correction for each kilometre south of datum
Free air correction	0.3 mgal/m	Add a correction for each elevation metre above sea-level
Bouguer correction	0.1 mgal/m	Subtract a correction for each metre above the datum station

The data for station A, the datum station, has been processed as below:

<i>Correction or calculation item</i>	<i>Station A (datum)</i>
Latitude correction	$-(0 \times 0.8)$
Free air correction	$+(50 \times 0.3)$
Bouguer correction	$-(0 \times 0.1)$
Total correction	+15.0
Observed g at station	979158.1
Corrected g at station	979173.1
Corrected g at datum station	979173.1
Gravity anomaly at station (corrected g – corrected datum g)	0

- (i) Outline the role of the Bouguer correction in the reduction of gravity data. 1
- (ii) Using a table format and the information provided, reduce the gravity data for stations B and C, and draw an inference about the geology of the survey data. 4

Question 28 continues on page 32

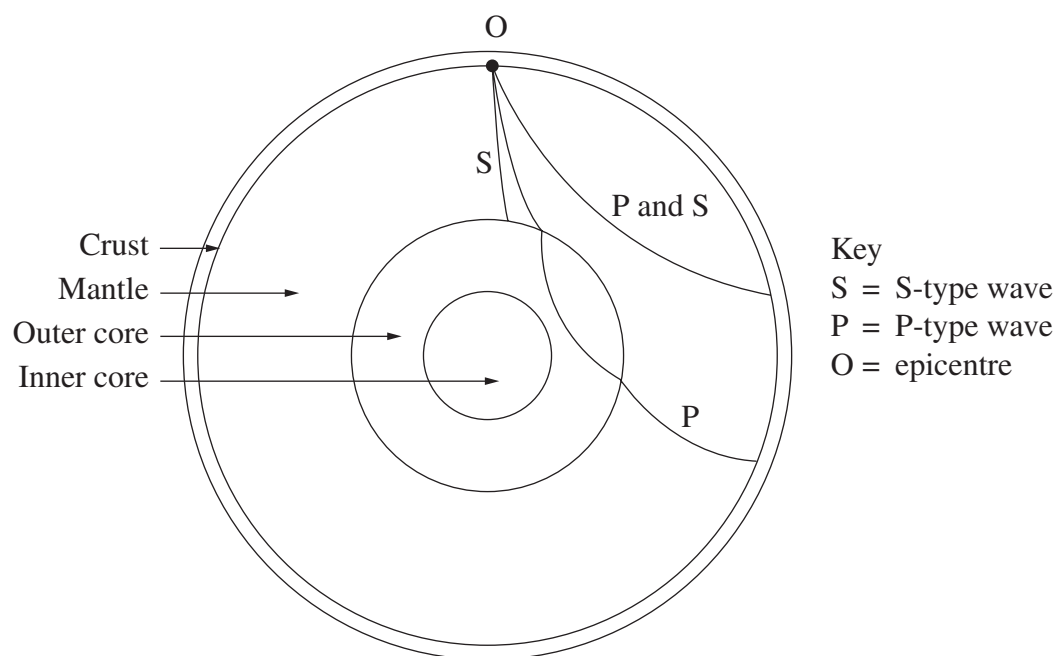
Question 28 (continued)

- (b) (i) During your study of geophysics you carried out a first-hand investigation to model the principles of reflection and refraction of seismic waves. 2

Describe how you ensured that the data obtained in your investigation were reliable.

- (ii) The diagram shows the path of seismic waves through Earth. 4

Account for the curved path followed by all the waves, and justify why the S-type wave stops at the outer core and the P-type wave does not.



- (c) 'Our understanding of Earth has changed as a result of developments in geophysics.' 7

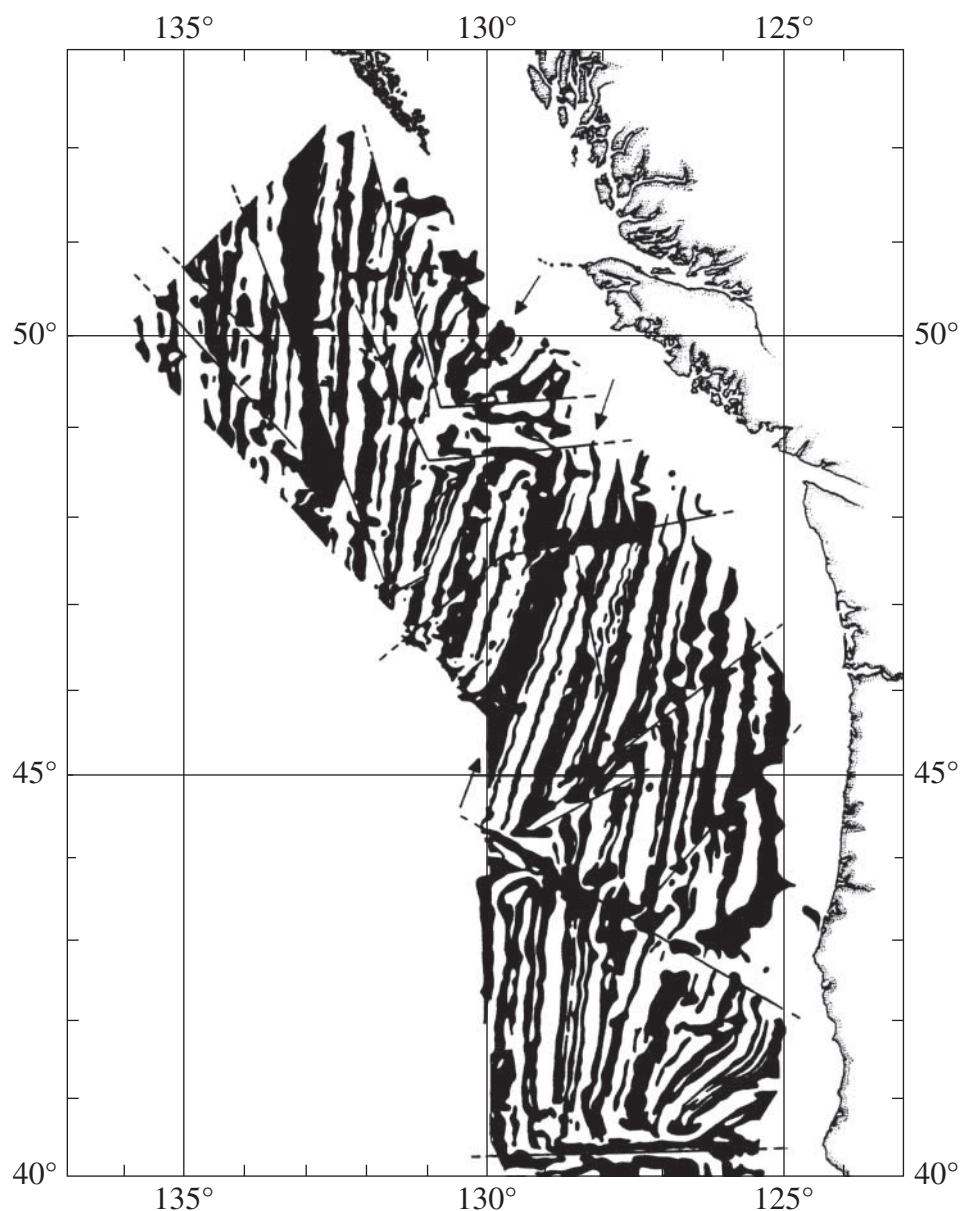
Discuss this statement with reference to the geophysical methods you have studied.

Question 28 continues on page 33

Question 28 (continued)

- (d) (i) The map below shows a pattern of magnetic anomalies recorded in the rocks of the seabed. The stripes represent areas of positive and negative magnetic anomaly.

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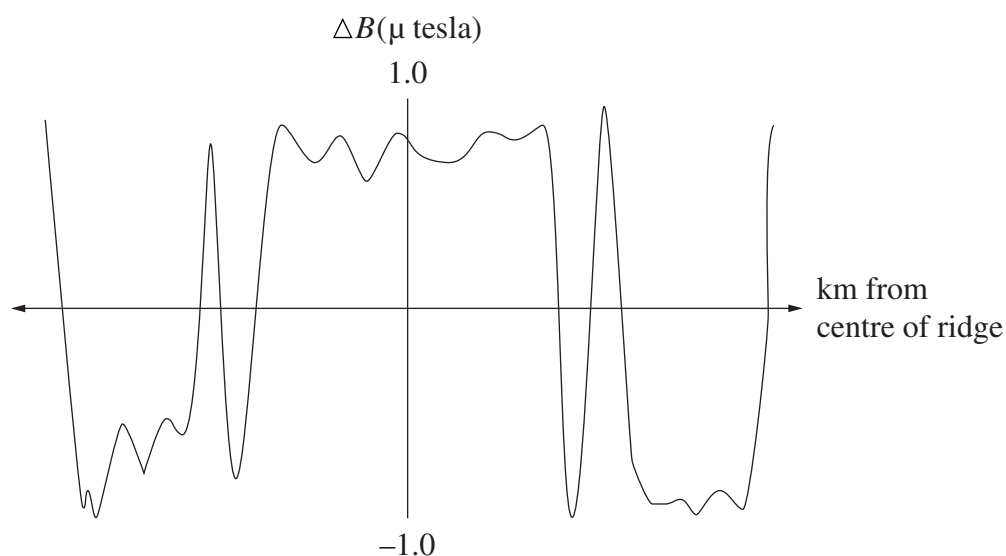
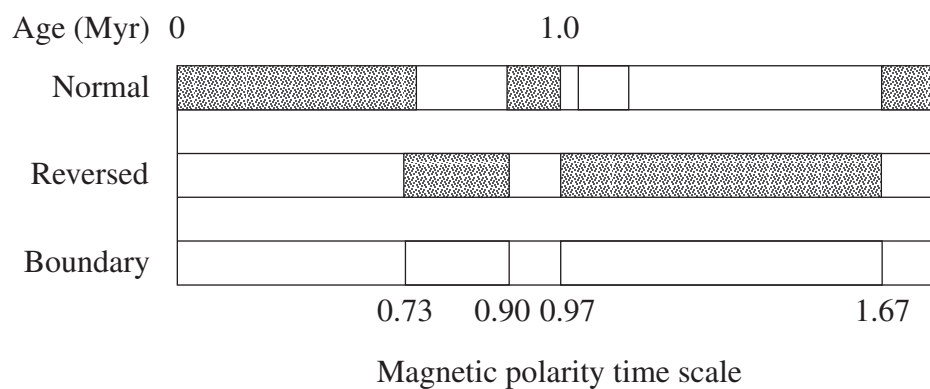
Explain the formation of these paleomagnetic patterns.

Question 28 continues on page 34

Question 28 (continued)

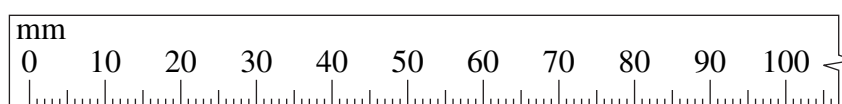
- (ii) The spreading rate of an ocean can be calculated using paleomagnetic data as shown. 3

Use the information provided to calculate the spreading rate of the ocean, in units of mm/yr.



Magnetic anomaly profile across the mid-ocean ridge

This diagram is to scale.



1 mm = 1 km

End of Question 28

Question 29 — Medical Physics (25 marks)

- (a) The acoustic impedance and density of a number of different types of body tissue, ultrasound gel and air are shown in the table:

<i>Material</i>	<i>Acoustic impedance</i> ($\text{kg m}^{-2} \text{s}^{-1} \times 10^6$)	<i>Density</i> (kg m^{-3})
Fat	1.38	9.25×10^2
Skin	1.52	1.00×10^3
Ultrasound gel	1.54	1.01×10^3
Air	0.0004	1.3

- (i) Calculate the velocity of sound in fat tissue. **2**
- (ii) Ultrasound gel is used to overcome the excessive reflection from the skin during a scan. **3**

Explain why this is necessary, justifying your response with calculations.

- (b) During your study of Medical Physics you identified data sources, and gathered, processed and presented information to explain why MRI scans can be used to detect abnormalities in the body.
- (i) Describe the criteria you would use to determine the reliability of a data source for this purpose. **3**
- (ii) Explain why MRI scans can be used to detect cancerous tissues. **3**
- (c) Advances in our understanding of the electromagnetic spectrum have allowed scientists to investigate the human body in more detail. **7**

Assess the impact of these advances on the development of medical technologies.

Question 29 continues on page 36

Question 29 (continued)

- (d) Bones can be viewed with a range of different medical imaging techniques. The images shown below were obtained using three different techniques.

*A**B**C*

- (i) The images shown above are an X-ray, a CAT scan and a bone scan (in no particular order). **1**

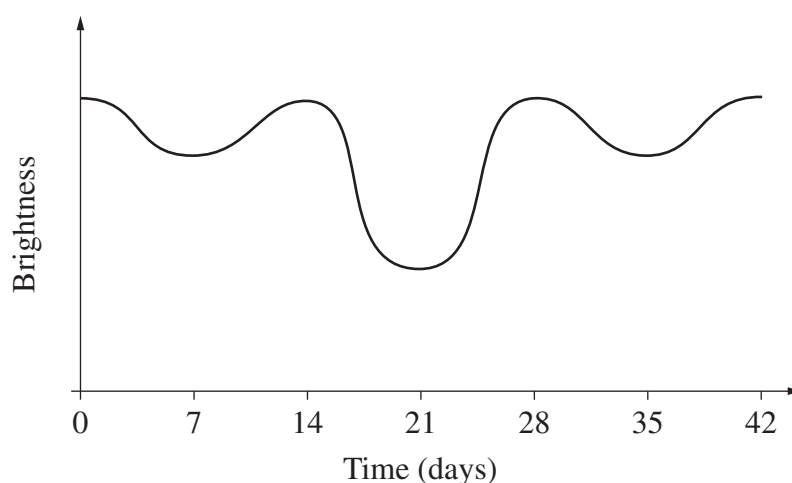
Identify the images labelled *A*, *B* and *C*.

- (ii) Compare the advantages and disadvantages of CAT scans to X-ray images. **3**
- (iii) Contrast the information provided by bone scans with that obtained by CAT scans and X-ray images. **3**

End of Question 29

Question 30 — Astrophysics (25 marks)

- (a) (i) Describe the spectroscopic observations that would determine whether a particular star is really a binary star system. **2**
- (ii) The graph represents the variation in brightness of a binary star system. **3**



Given that the mass of the system is determined to be 6×10^{32} kg, calculate the average distance between the stars within the system.

- (b) During your study of Astrophysics you performed a first-hand investigation into the spectra produced by different objects under different conditions.
- (i) Explain how you determined that the data you obtained were reliable. **2**
- (ii) Explain how the absorption spectrum of a star is produced, and how it can be used to determine the star's composition. **4**
- (c) Astronomers employ a range of instruments and techniques to observe celestial objects. **7**

Assess the impact of technological advances on our understanding of the cosmos.

Question 30 continues on page 38

Question 30 (continued)

- (d) The Hertsprung-Russell (H-R) diagram depicts a possible life cycle path of a known star.



- | | |
|---|----------|
| (i) Describe the reactions that occur in stars at the points marked A , B and C in its life cycle. | 3 |
| (ii) Explain what type and mass of star is most likely to be formed at point A . | 2 |
| (iii) Compare the life cycle of a star that has a mass greater than 10 solar masses with the one depicted at point A . | 2 |

End of Question 30

Question 31 — From Quanta to Quarks (25 marks)

- (a) The Bohr picture of the atom explains the energy of the photons emitted when an electron falls from an initial orbit n_i to find an orbit of n_f .

The various energies depend upon the quantity $\left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$ and the values for this term are given in the table.

Table of values of $\left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$ for values of n_i and n_f from 1 to 6

$n_f \backslash n_i$	1	2	3	4	5	6
1	—	0.7500	0.8889	0.9375	0.9600	0.9722
2	—	—	0.1389	0.1875	0.2100	0.2222
3	—	—	—	0.0486	0.0711	0.0833
4	—	—	—	—	0.0225	0.0347
5	—	—	—	—	—	0.0122
6	—	—	—	—	—	—

- (i) Identify the physical reason for about one-half of the table appearing blank. **1**
- (ii) Calculate the energy of the photon emitted when an electron falls from $n_i = 4$ to $n_f = 3$. **4**
- (b) You have gathered, processed and analysed information related to the development of atomic theory.
- (i) Describe how you ensured that the information you gathered was reliable. **2**
- (ii) The atomic theory changed as a result of the contributions of both Heisenberg and Pauli. **4**

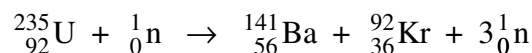
Analyse how the work of both scientists modified the atomic theory at that time.

Question 31 continues on page 40

Question 31 (continued)

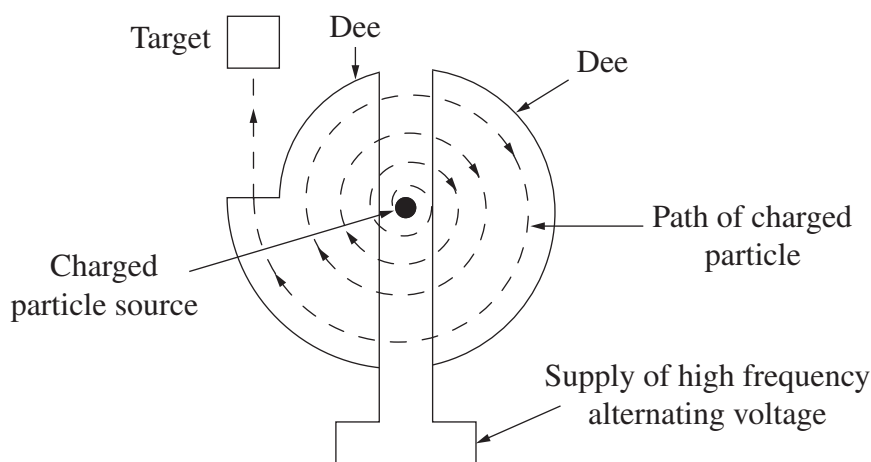
- (c) Australia has a large supply of uranium which may be used in fission reactors to create energy. The equation describes the relevant transmutation reaction:

7



Analyse how the process described in this equation has been developed into a technology which produces a sustained and controlled amount of energy.

- (d) The cyclotron, invented in 1932, accelerates charged particles to a very high speed. The diagram shows the basic design of a cyclotron. The Dees provide a strong magnetic field into the plane of the page.



- (i) Explain the physical principles involved in the design of the cyclotron. 3
- (ii) Account for the use of the cyclotron (or other accelerator) in the development of our understanding of matter. 2
- (iii) Quarks are an important part of the Standard Model of Matter. The table shows the six types of quark and their charge (in units of e , the charge on an electron). 2

Quark	Up	Down	Strange	Charm	Bottom	Top
Charge	$+\frac{2}{3}e$	$-\frac{1}{3}e$	$-\frac{1}{3}e$	$+\frac{2}{3}e$	$-\frac{1}{3}e$	$+\frac{2}{3}e$

Identify the quark composition of the proton and the neutron.

End of Question 31

Question 32 — The Age of Silicon (25 marks)

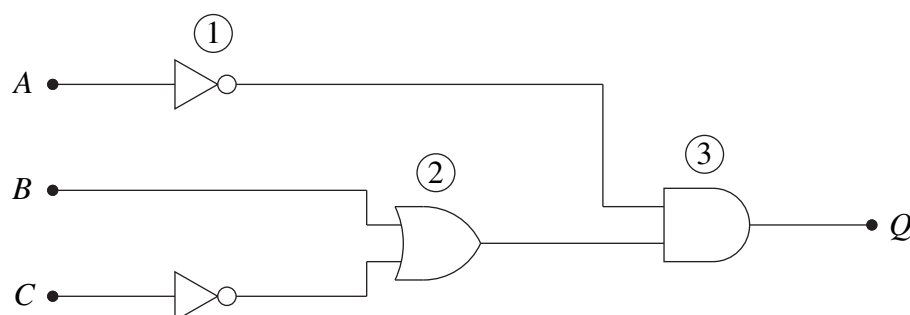
- (a) Lately, traffic control authorities in NSW have been replacing the incandescent bulbs in traffic lights with arrays of light emitting diodes (LEDs).

- (i) Describe the structure and operation of an LED. 2
- (ii) Explain why LEDs are preferable to ordinary light bulbs when used in traffic lights. 3

- (b) (i) During your study of the Age of Silicon you identified data sources, and gathered, processed and analysed information to outline the rapid development of electronics. 3

Describe the criteria you would use to determine the reliability of a data source for this purpose.

- (ii) Circuits containing logic gates are an integral part of modern electronics. An example of such a circuit is given below. 3



Identify each gate labelled ①, ② and ③ and, using a truth table, determine the value of Q when $A = 0$, $B = 1$ and $C = 1$.

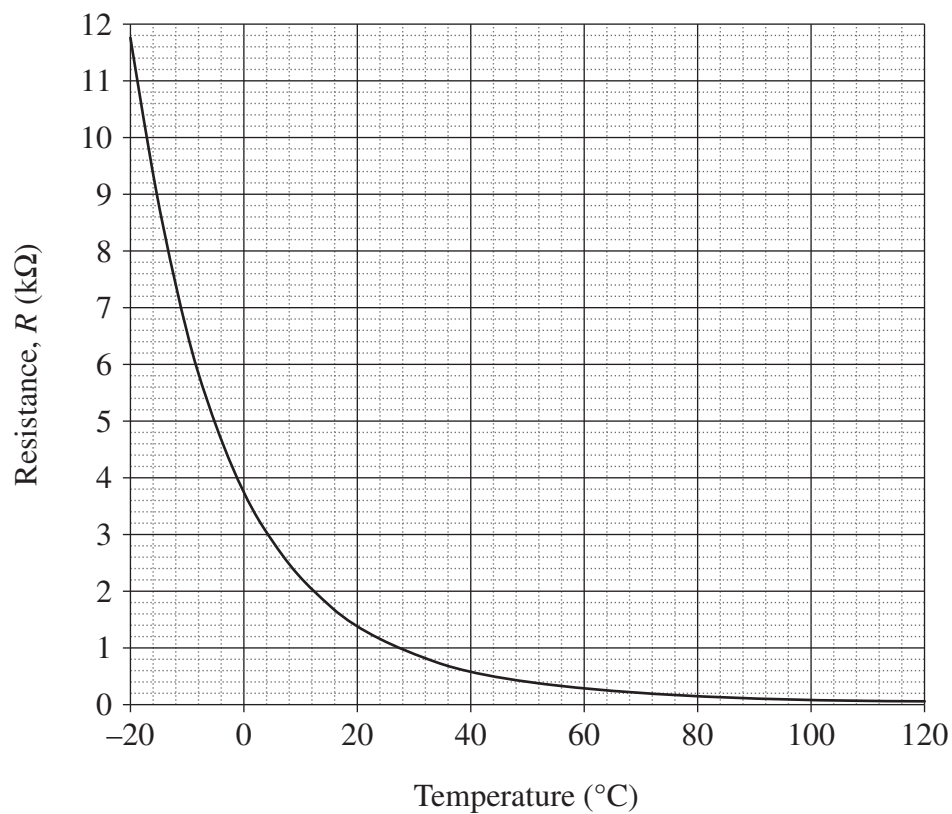
- (c) Computer developments since World War II (1945) have been characterised by increasing performance at decreasing cost. This trend may or may not continue in the future. 7

Assess this statement.

Question 32 continues on page 42

Question 32 (continued)

- (d) The resistance of a thermistor as a function of temperature is as shown.



- (i) Deduce the sign of the temperature coefficient of the thermistor. Explain your answer.

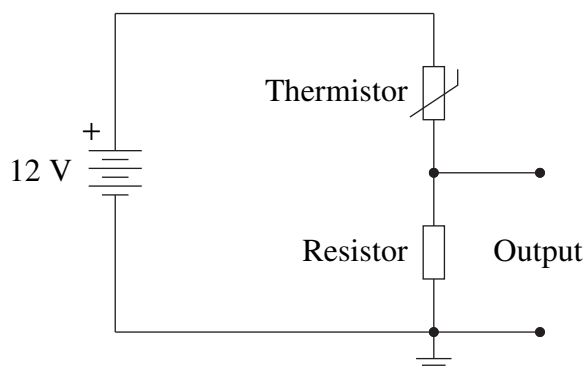
1

Question 32 continues on page 43

Question 32 (continued)

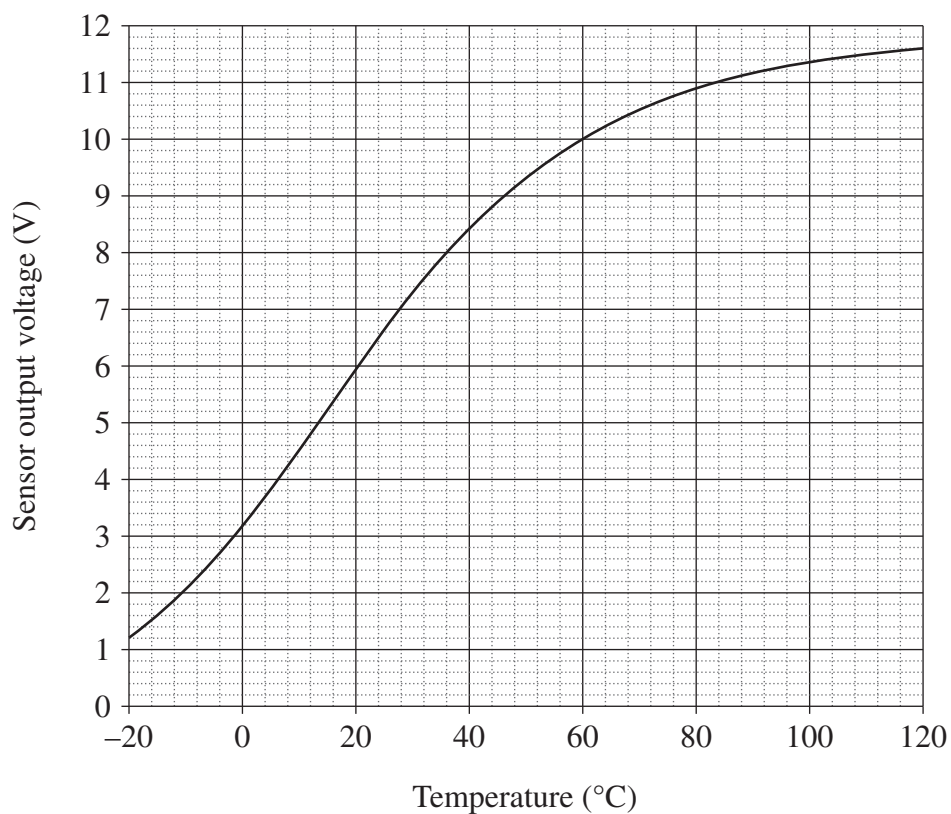
- (ii) A potential divider sensor circuit using this thermistor and a fixed resistor is built as shown.

2



A chamber temperature of $+20^{\circ}\text{C}$ is required. On connecting a resistor, the sensor output voltage as a function of temperature is measured and plotted as below.

Determine the approximate sensitivity of the potential divider sensor in the temperature range 6°C to 20°C .



Question 32 continues on page 44

Question 32 (continued)

- (iii) The output of the potential divider sensor is to be converted to give a **4**
–10.0 V signal at 20°C.

Design a suitable feedback amplifier to achieve this conversion. Base your circuit on an operational amplifier and assume a ± 15 V power supply. Show, using a fully labelled diagram, how your circuit should be connected to the temperature sensor.

End of paper

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \quad \text{therefore} \quad a_{\text{av}} = \frac{v-u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen	KEY																2 He 4.003 Helium					
3 Li 6.941 Lithium	4 Be 9.012 Beryllium	Atomic Number		79 Au 197.0 Gold		Symbol of element		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon				
		Atomic Weight		Gold		Name of element		Aluminium		Silicon		Phosphorus		Sulfur		Chlorine						
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium																	18 Ar 39.95 Argon				
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel			29 Cu 63.55 Copper		30 Zn 65.41 Zinc	
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.94 Molybdenum		43 Tc [97.91] Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		54 Xe 131.3 Xenon
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57–71 Lanthanides		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.8 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		
87 Fr [223.0] Francium	88 Ra [226.0] Radium	89–103 Actinides		104 Rf [261.1] Rutherfordium		105 Db [262.1] Dubnium		106 Sg [266.1] Seaborgium		107 Bh [264.1] Bohrium		108 Hs [277] Hassium		109 Mt [268] Meitnerium		110 Ds [271] Darmstadtium		111 Rg [272] Roentgenium				

Lanthanides

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
138.9	140.1	140.9	144.2	[144.9]	150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium

Actinides

89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
[227.0]	232.0	231.0	238.0	[237.0]	[244.1]	[243.1]	[247.1]	[247.1]	[251.1]	[252.1]	[257.1]	[258.1]	[259.1]	[262.1]
Actinium	Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.

The atomic weights of Np and Tc are given for the isotopes ^{237}Np and ^{99}Tc .

2006 HSC Notes from the Marking Centre Physics

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2006 HSC NOTES FROM THE MARKING CENTRE

PHYSICS

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Physics. It provides comments with regard to responses to some questions in the 2006 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section.

This document should be read along with the relevant syllabus, the 2006 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Physics.

General Comments

In 2006, 9116 candidates attempted the Physics examination. The most popular options were From Quanta to Quarks (43.5%) and Medical Physics (27%).

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course, including the Prescribed Focus Areas. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. It is important to understand that the Preliminary HSC course is assumed knowledge for the HSC course.

In 2006, at least one question in Section I Part B and one part of the Section II option questions focused on the mandatory skills content in Module 9.1. Candidates who had actively planned and performed practical experiences clearly demonstrated a deeper knowledge and understanding of the content described in this module.

Overall, the level of understanding of Physics concepts indicated by the responses was appropriate for most HSC candidates. Candidates need to be reminded that the answer space provided and the marks allocated are guides to the maximum length of response required. Similarly, the key word used in the question gives an indication of the depth of the required response. Candidates should use examination time to analyse the question and plan responses carefully, working within that framework to produce clear and concise responses. Responses may include the use of dot points, diagrams and/or tables, and should avoid internal contradictions. This is particularly so in holistic questions which need to be logical and well-structured. There was evidence that some candidates had a very poor knowledge of basic definitions specific to terminology associated with the course.

Better responses indicate that candidates had followed the instructions provided on the examination paper. In these responses candidates:

- set out all working for numerical questions
- thought carefully about the units to be used and the quantities to be substituted into formulae

- did not repeat the question as part of the response
- looked at the structure of the whole question and noted that in some questions the parts follow from each other ie responses in part (a) lead to the required response in part (b) etc
- used appropriate equipment, for example, pencils and a ruler to draw diagrams and graphs. (A clear plastic ruler helps candidates to plot points that are further from the axes and rule straight lines of best fit.)

In Section II the option question is divided into a number of parts. Candidates should clearly label each part of the question when writing in their answer booklets. In part (c) of the 2006 option questions the best responses presented ideas coherently and included the correct use of scientific principles and ideas. Many candidates wrote a lot of information that was not relevant to the question. Some responses showed evidence of rote learning of an anticipated answer based on a single source. These responses did not address the syllabus content and/or outcomes being assessed and hence did not score full marks. Candidates are required to attempt one question only in Section II, but some candidates responded to more than one option question. Candidates are strongly advised to answer the option they have studied in class.

Section I – Core

Part A – Multiple choice

Question	Correct Response
1	C
2	B
3	D
4	B
5	D
6	A
7	D
8	A

Question	Correct Response
9	B
10	A
11	A
12	C
13	D
14	C
15	A

Part B

Question 16

- Better responses displayed recognition of the vector nature of velocity and gave both magnitude and direction in the answer.
- Better responses used Pythagoras' theorem to calculate the initial velocity.
- Better responses used the relevant equation, correctly substituted data, and identified the directions of the vector quantities involved.

Question 17

Better responses focused on the forces on the spacecraft rather than on the astronauts and provided the analysis of two parts – geostationary orbit and re-entry. These responses clearly indicated that the centripetal force needed to maintain a stable orbit is provided by the gravitational force between the earth and the spacecraft and that the two forces do not cancel each other out. Better responses also described some of the forces involved in re-entry and explained their effects on the motion of the spacecraft.

Sample response:

During stationary orbit, the spacecraft travels at a constant speed in orbit around the Earth. The Earth's gravity provides the centripetal force for the spacecraft's orbit. In order to return to Earth, the spacecraft must decelerate by firing its boosters in the opposite direction of motion. This creates a force to slow the speed of the spacecraft and hence reduce its energy. When the energy is reduced, its radius of orbit must also decrease since the two are related so the spacecraft descends. The particles in the upper atmosphere provide a friction force opposite to the direction of motion of the craft, slowing it down further which lowers its orbit and increases the friction. Gravity pulls the spacecraft back to Earth where the craft can land.

Question 18

Better responses recognised the inverse relationship between E_p and r in the formula, and used ratios to logically determine the absolute value of E_{p3} and subtract it from 1MJ.

Question 19

The better responses described the motor effect clearly, and then outlined the effect and/or relationship of each component in the diagram on the galvanometer in a logical and concise way.

Question 20

The best responses plotted the data correctly, extracted data using the line of best fit, and demonstrated a clear understanding of the forces involved as well as their relationship to the graphical data.

Question 21

Most candidates were able to describe the impacts of identified applications on society and the environment. In the better responses, candidates indicated that they understood the significance of the word 'potential' and also made a clear judgement of the impacts. The more concise of these responses took less space than that provided in the question paper. A common misconception was that the Meisner effect is related to the use of superconductors in Maglev trains.

The following examples illustrate some of the different ways in which students structured their responses.

Sample response 1:

	<i>Applications</i>	<i>Advantages</i>	<i>Disadvantages</i>
<i>Maglev Trains</i>	<i>Society</i>	frictionless form of transport allowing super fast speeds	- the costs of implementation are detrimental - maintenance is a problem
	<i>Environment</i>	does not use fossil fuels to provide transport	the superconductors must be kept at a critical temperature which is difficult to maintain and uses a large amount of energy
<i>Energy Supply</i>	<i>Society</i>	allows for large scale distribution of energy without power loss as there is no resistance in cables	- replacement of manual labour causing unemployment - lack of technology to maintain a large distribution grid
	<i>Environment</i>	power lines are unaesthetic	- large amounts of energy used to keep cables at critical temperatures - costs and wastage of materials in creating cables for large scale supply

Despite the obvious benefit of using superconductors because of their magnetic and conducting properties, potential applications are prohibited by the costs involved in implementation. Also, we lack the technology to implement all potential applications, however it may be possible in the future. Overall the large scale use of Maglev trains and superconducting power grids is advantageous to both society and the environment.

Sample response 2:

<i>Application of superconductor</i>	<i>Impact on Society</i>	<i>Impact on environment</i>	<i>Assessment</i>
<i>Maglev trains</i>	- Provides faster, more energy-efficient transport - More expensive to build - i.e. higher fares	- Less use of fossil fuels to power train, therefore less random emissions - Less coal needs to be mined for power stations	<i>Beneficial to both society and environment – however is costly</i>
<i>Transporting electricity</i>	- More efficient transportation - Zero power losses - Safer as DC is used - No need to have expensive transformers - Cheaper to transport - Smaller cables means more aesthetic benefits	- Less fossil fuels emitted into atmosphere – therefore, less air pollution - Reduced likelihood of acid rain	<i>Beneficial to society as a cheaper, more efficient transportation takes place, less harm to society in the form of pollution</i>
<i>Superconducting generation of power</i>	- More efficient power production - No need for AC and transformers - Cheaper electricity - Cleaner energy	No fossil fuels used for power generation	<i>Less environmental impacts, society gets cheaper energy</i>

Sample response 3:*Impacts on society and the environment*

	<i>Positive</i>	<i>Negative</i>
<i>Society</i>	<i>Faster transit eg Maglev trains Scientific research – more powerful synchotrons</i>	
<i>Environment</i>	<i>Decreased fossil fuel dependence from for example Maglev trains. Less voltage required for a good electromagnet so less power needs to be generated, so less fossil fuels burnt.</i>	<i>More cooling, and therefore more power needed in many cases – for example many modern superconductors must be cooled with liquid helium. Fossil fuels are burnt to manufacture this.</i>

From the table, it is clear that overall, the potential applications of superconductors have a positive impact on society and the environment.

Question 22

- (a) In most responses an attempt was made to include a ‘cause and effect’ relationship between the motion of the magnet and induced currents. In better responses there was a clear link between the resultant force and the interaction of the magnetic fields.
- (b) A number of responses incorrectly identified that -50°C was below the critical temperature of copper and incorrectly responded in terms of superconductivity.

Question 23

- (a) Better responses showed the conductor with overlapping bands, using either labels or shading to show this, as well as indicating the appropriate energy gaps in semiconductors and insulators.
- (b) Some better responses used tables to answer this part of the question.

Sample response:

<i>Insulators</i>	<i>Semiconductors</i>	<i>Conductors</i>
<i>High electrical resistance as large amounts of energy are required to move electrons into conduction band, shown in large gap.</i>	<i>Medium resistance as the energy gap is smaller and therefore less energy needed to move electrons across it.</i>	<i>Low resistance – in diagram it shows that the valence band is the same as the conduction band.</i>

- (c) Better responses gave concise explanations of the relationship between the change in R and the dopant without referring to temperature change. Wordy responses often led to contradictions within the response.

Question 24

Better responses included more than one origin of unwanted heat production and linked a method of overcoming the problem with each of the sources of the heat production.

Sample response:

In transformers the soft iron core can have eddy currents induced in it due to the changing magnetic fields, producing heat. This is overcome through using laminations in between layers of soft iron in the core, making eddy currents. The resistance within the wires of the coils can also produce heat. This is overcome by using thicker wires (although this can increase current) or a coolant such as oil or water.

Question 25

- (a) Better responses clearly indicated the main features of the role of the deflecting plates and electrodes in the electron gun separately and provided additional information to that available from the stimulus material.
- (c) The better responses manipulated the equation to provide a correct interpretation about the frame of reference.

Section II – Options

Question 28

- (d) (ii) The majority of candidates who correctly applied the scale to determine distance and time values were able to establish the correct sea floor spreading rate.

Question 29

- (a) (i) In the better responses, the calculations included the order of magnitude for acoustic impedance provided in the heading of the table.
- (a) (ii) Better responses identified the significance of the similarity in acoustic impedance between gel and skin, calculated I_r/I_o values for both interfaces and used the results of the calculation to describe the amount of ultrasound that penetrated into the body.
- (b) (ii) Better responses identified the importance of increased water in cancerous tissue and described how MRI detects the presence of hydrogen nuclei.
- (c) Better responses had a value statement affirming the positive effects of our understanding of the electromagnetic spectrum on the development of medical imaging technologies and related techniques together with a statement connecting each particular technique with the part of the electromagnetic spectrum upon which it relied.
- (d) (ii) In some of the better responses information was represented in a table format demonstrating a clear comparison and understanding of the two techniques.
- (d) (iii) Better responses identified the differences in the three techniques and the resulting differences in the data obtained. Representing the differences in a table format provided responses that were clear and unambiguous.

Question 30

- (a) (i) Better responses correctly identified Doppler shifts involving two stars as occurring simultaneously and periodically.
- (a) (ii) Better responses calculated the required distance by correctly identifying the period of the binary from the graph and also by using correct time units.
- (b) (ii) Better responses described the appearance of the absorption spectrum of a star, in terms of dark spectral lines upon a continuum spectrum. The best responses explained their occurrence in terms of atomic electrons absorbing specific photons from the continuum spectrum emitted by the stellar core in order to ‘jump’ to higher-energy orbits.
- (c) Most responses were able to give some technological advances and their benefits. Better responses clearly explained how these benefits lead to a better understanding of the cosmos.

Many responses demonstrated difficulty in relating the technological advances to an understanding of the cosmos suggesting that candidates are concentrating too much on individual ‘dot points’ and neglecting to synthesise the knowledge and understanding gained from the study of the option as a whole.

Question 31

- (a) (i) In the best responses, candidates demonstrated an understanding of the information given at the beginning of the question regarding the energy of the photons to give a physical reason for the table being blank.
- (ii) Better responses showed working that followed a logical sequence of several calculations to arrive at the correct answer.
- (b) (ii) Better responses showed evidence of planning.
- (c) The best responses were able to identify the source of nuclear energy in the mass defect from the equation and related this to the technology of the nuclear fission reactor to control the neutron numbers in the resulting chain reaction.

Most responses showed some understanding of the structure of the nuclear reactor in its role of providing useable thermal energy.

- (d) (i) Better responses indicated a familiarity with the operation of the cyclotron and correctly identified that the alternating current caused the increase in speed of the charged particle and the magnetic field caused centripetal force on the particle.
- (d) (ii) Better responses gave clear reasons to show how the cyclotron has added to our understanding of matter.

Question 32

- (b) (i) Better responses provided evidence that candidates had carefully read the question and had structured their responses accordingly.
- (b) (ii) Better responses clearly identified the resulting changes after each gate in the truth table.
- (d) (ii) Better responses demonstrated that data had been accurately read from the graph.
- (iii) Better responses demonstrated an understanding of the structure and function of a feedback amplifier using an operational amplifier as well as demonstrating the way this was to be linked to other devices.

Physics

2006 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.2.1.2.1, 9.2.1.2.3	H6, H7
2	1	9.2.2.2.8, 14.1 (a)	H6, H14
3	1	9.2.4.2.3, 9.2.4.3.1	H6
4	1	9.2.2.2.1, 14.1 (a)	H6, H9, H14
5	1	9.2.2.3.5, 9.2.2.2.10, 12.4 (b)	H6, H12
6	1	9.3.1.2.5	H9
7	1	9.1.2.1, 9.3.1.3.3, 12.4 (b)	H7, H9, H12
8	1	9.3.2.2.3, 14.1 (a)	H9, H14
9	1	9.3.3.2.1	H7
10	1	9.3.4.2.2, 14.3 (d)	H7, H14
11	1	9.4.4.2.1	H8, H10
12	1	9.4.1.2.3	H9
13	1	9.4.4.2.3	H7, H10
14	1	9.4.1.2.7, 9.4.1.3.3, 12.4 (b)	H7, H12
15	1	9.4.2.2.1, 12.3 (c)	H10, H12
Section I Part B			
16 (a)	1	9.2.2.3.1	H6, H9
16 (b)	1	9.2.2.3.2, 12.4 (b)	H6, H9, H12
16 (c)	2	9.2.2.3.2, 12.4 (b)	H6, H9, H12
16 (d)	2	9.2.2.3.2, 12.4 (b)	H6, H9, H12
17	6	8.4.2.2.10, 9.2.2.2.8, 9.2.2.2.9	H7, H9
18	3	9.2.1.2.2, 9.2.1.2.3, 12.4 (b)	H6, H12
19	3	9.3.1.2.4, 9.3.1.3.5	H7, H9

Question	Marks	Content	Syllabus outcomes
20 (a)	2	9.3.1.2.2, 12.3 (c)	H9, H12
20 (b)	2	9.3.1.2.2, 13.1 (f), (g)	H9, H13
20 (c)	1	9.3.1.2.4, 12.3 (c)	H9, H12
20 (d)	3	9.3.1.2.5, 12.4 (d)	H9, H12
21	6	9.4.4.2.7, 9.4.4.3.5, 9.4.3	H3, H4
22 (a)	3	9.3.2.2.5	H7, H9
22 (b)	2	9.3.2.2.7, 9.4.4.2.5, 14.1 (d)	H7, H9, H14
23 (a)	2	9.4.3.2.2, 13.1 (e)	H9, H13
23 (b)	2	9.4.3.2.2	H9
23 (c)	2	9.4.3.2.6	H9
24	3	9.3.4.3.3, 9.3.4.3.4	H6, H7, H9
25 (a)	2	9.4.1.2.9	H7, H9
25 (b)	2	9.2.4.3.5, 12.4 (b)	H6, H12
25 (c)	2	9.2.4.3.5, 12.4 (b)	H6, H12
26	4	9.4.2.2.3, 9.4.2.2.4, 9.4.2	H2, H8, H9
27 (a)	2	9.4.1.3.2, 11.3 (b)	H10
27 (b)	2	9.4.1.2.1, 9.4.1.3.2, 11.3 (b)	H10, H11
Section II			
Question 28 — Geophysics			
28 (a) (i)	1	9.5.2.2.6, 9.5.2.2.7	H9
28 (a) (ii)	4	9.5.2.2.8, 9.5.2.3.5, 12.3 (c), 13.1 (e)	H12, H13
28 (b) (i)	2	9.5.3, 11.2 (d)	H11
28 (b) (ii)	4	9.5.3.2.2, 9.5.3.2.3, 14.1 (a)	H8, H14
28 (c)	7	9.5.1.2.2, 9.5.2.2.1, 9.5.2.2.9, 9.5.3.2.5, 9.5.3.2.7, 9.5.4.2.2, 14.3 (b)	H1, H2, H3, H14
28 (d) (i)	4	9.5.4.2.1, 9.5.4.2.2, 9.5.4.2.3, 12.3 (c), 14.1 (c)	H9, H12, H14
28 (d) (ii)	3	9.5.4.3.2, 12.4 (b), 14.1 (a), (f)	H12, H14

Question	Marks	Content	Syllabus outcomes
Section II			
Question 29 — Medical Physics			
29 (a) (i)	2	9.6.1.3.5	H8
29 (a) (ii)	3	9.6.1.2.3, 9.6.1.2.4, 9.6.1.2.6, 12.4 (b)	H3, H12
29 (b) (i)	3	9.6.4.3.1, 9.6.4.3.2, 11.1 (a), 11.2 (d), 11.3 (c), 12.3 (a)	H11, H12
29 (b) (ii)	3	9.6.4.3.2, 9.6.4.3.3	H8, H9
29 (c)	7	9.6.2, 9.6.3.2.5, 9.6.4.3.5, 14.3 (b)	H3, H9, H14
29 (d) (i)	1	9.6.2.3.4, 9.6.2.2.3, 12.3 (c)	H3, H12
29 (d) (ii)	3	9.6.2.2.4	H3, H10
29 (d) (iii)	3	9.6.2.3.2, 9.6.3.3.1, 14.1 (a)	H3, H14
Section II			
Question 30 — Astrophysics			
30 (a) (i)	2	9.7.5.2.1	H8
30 (a) (ii)	3	9.7.5.3.2	H8, H9
30 (b) (i)	2	9.7.3.3.1, 11.2 (d)	H11
30 (b) (ii)	4	9.7.3.2.5	H10
30 (c)	7	9.7.1, 9.7.3, 9.7.4, 14.3 (b)	H3, H10, H14
30 (d) (i)	3	9.7.6.2.3, 9.7.6.3.2	H9
30 (d) (ii)	2	9.7.6.3.2, 12.3 (c)	H9, H12
30 (d) (iii)	2	9.7.6.2.2, 14.1 (a), 14.3 (c)	H9, H14
Section II			
Question 31 — From Quanta to Quarks			
31 (a) (i)	1	9.8.1.2.2	H9, H10
31 (a) (ii)	4	9.8.1.3.3, 12.4 (d), (e)	H9, H10, H12
31 (b) (i)	2	9.8.2.3.2, 11.1 (a), (c), 12.3 (a), 12.4 (b)	H11, H12
31 (b) (ii)	4	9.8.2.3.2, 9.3.3.2.6	H1, H2, H10
31 (c)	7	9.8.3.2.3, 9.8.3.2.9, 9.8.3.3.2, 9.8.3.2.11, 9.8.4.2.1, 14.3.(b)	H4, H7, H14
31 (d) (i)	3	9.8.4.2.4, 12.3 (c), 14.1 (g)	H6, H7, H9, H12, H14
31 (d) (ii)	2	9.8.4.2.4	H6, H9

Question	Marks	Content	Syllabus outcomes
31 (d) (iii)	2	9.8.4.2.5, 12.3 (c)	H10, H12
Section II			
Question 32 — The Age of Silicon			
32 (a) (i)	2	9.9.4.2.3	H7
32 (a) (ii)	3	9.9.4.2.3, 9.9.4.3.3, 12.4 (b)	H7, H12
32 (b) (i)	3	9.9.5.3.2, 11.1 (a), 11.2 (d), 11.3 (c), 12.3 (a)	H11, H12
32 (b) (ii)	4	9.9.5.3.2	H12
32 (c)	7	9.9.1.2.2, 9.9.1.2.3, 9.9.1.3.1, 9.7, 14.3 (b)	H5, H7, H14
32 (d) (i)	1	9.9.3.2.5, 12.3 (c)	H7, H12
32 (d) (ii)	2	9.9.3.2.4, 12.3	H7, H12
32 (d) (iii)	4	9.9.6.2.3, 9.9.6.2.4, 9.9.6.3.3, 13.1 (e)	H7, H13

2006 HSC Physics Marking Guidelines

Section I, Part B

Question 16 (a)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
• Gives correct answer	1

Question 16 (b)

Outcomes assessed: H6, H9, H12

MARKING GUIDELINES

Criteria	Marks
• Correctly determines magnitude of initial velocity	1

Question 16 (c)

Outcomes assessed: H6, H9, H12

MARKING GUIDELINES

Criteria	Marks
Correctly calculates maximum height	2
- Identifies the equation(s) required but uses them incorrectly OR - Calculates time to reach maximum height, but does not correctly determine the height	1

Question 16 (d)

Outcomes assessed: H6, H9, H12

MARKING GUIDELINES

Criteria	Marks
Correctly determines the range	2
- Correctly determines time of flight but not the range OR - Identifies the equation(s) required to produce the correct answer	1

Question 17

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
Describes forces both during stable orbit and upon re-entry, and relates these to the result	5–6
Describes forces either during stable orbit or upon re-entry, and relates to the result	3–4
Identifies forces both during stable orbit and upon re-entry but does not relate them to the result	2
Identifies one of the forces during stable orbit or upon re-entry	1

Question 18

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
• Demonstrates appropriate problem solving techniques to completion	3
• Calculates GPE at any of satellite positions OR • Calculates value of Gm_1m_2	2
• Identifies correct equation OR • Recognises $W = \Delta GPE$	1

Question 19

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
• Describes how the components interact to enable the meter to operate including reference to the motor effect	3
• Correctly describes purpose of some components and origin of motor effect, but does not discuss radial field, constant torque, or linear scale	2
• Describes function of only ONE component of galvanometer	1

Question 20 (a)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
• Identifies current directions and justifies answer	2
• Identifies directions but does not justify OR • Explains observations but chooses the incorrect direction of current flow	1

Question 20 (b)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly plots data and draws an appropriate trend line/line of best fit	2
• Correctly plots data but does not draw a trend line/line of best fit OR • Incorrectly plots data but draws a plausible trend line/line of best fit	1

Question 20 (c)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
• Correctly finds mass	1

Question 20 (d)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
• Correctly determines slope of trend line or uses an appropriate data point, identifies force relationship formula, substitutes constant values, and determines value of separation	3
• Identifies force relationship formula and determines slope of trend line but goes no further	2
• Identifies force relationship formula	1

Question 21

Outcomes assessed: H3, H4

MARKING GUIDELINES

Criteria	Marks
- Identifies at least two applications. For at least two describes at least one impact on society and/or the environment - Makes a value statement with respect to at least two impacts identified	5–6
Identifies at least two applications. For at least two describes an impact on society and/or the environment	3–4
- Identifies at least two applications OR - Identifies one application and an impact	2
- Identifies an application OR - Identifies an impact	1

Question 22 (a)

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
Explains the creation of eddy currents and their effects	3
Identifies the presence of eddy currents and their effect on the motion of the magnet	2
- Identifies eddy currents OR - Identifies Lenz's law OR - Attributes the motion solely to gravity	1

Question 22 (b)

Outcomes assessed: H7, H9, H14

MARKING GUIDELINES

Criteria	Marks
Identifies that the magnet slows down more than in a) due to a reduction in temperature causing a decrease in resistance of the copper, thus producing larger currents in eddy loops; the larger currents have a stronger magnetic field associated with them, causing the magnet to slow down more than at room temperature	2
Identifies that the magnet slows down more than in part (a)	1

Question 23 (a)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
Diagrams have conduction and valence bands, and a suitable gap between	2
- Diagrams illustrate the gap without mentioning bands OR - Indicates bands with incorrect gaps	1

Question 23 (b)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Refers to diagrams and correctly links to resistance for all three	2
- Refers to diagrams and correctly links to resistance for one or two of the cases OR - Describes generally the difference in R between conductors, semiconductors and insulators without reference to the diagram OR - Refers differences in diagrams without reference to conductors, semiconductors or insulators	1

Question 23 (c)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Identifies that changes occur when semiconductors are doped this way, and refers to difference in number of valence electrons between impurity and host material, relating this to a decrease in resistance	2
Identifies that doping changes the number of electrons or holes, but does not explain how this relates to resistance	1

Question 24

Outcomes assessed: H6, H7, H9

MARKING GUIDELINES

Criteria	Marks
Two valid strategies discussed	3
- Clearly links AC current and/or change in flux, eddy currents and heat OR - Links heat production to resistance to current flowing in the primary and secondary coils OR - Links eddy currents to heat and identifies two strategies OR - Links eddy currents to heat and describes a valid strategy	2
- States eddy current production OR - States heating in windings due to current flow OR - Outlines a strategy for heat reduction	1

Question 25 (a)

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
Correctly outlines roles of the deflection plates and the electrodes in the electron gun	2
Correctly outlines role of either the deflection plates or the electrodes in the electron gun	1

Question 25 (b)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies and substitutes data into equation	2
• Identifies correct equation	1

Question 25 (c)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies and substitutes data into equation	2
• Identifies correct equation	1

Question 26

Outcomes assessed: H2, H8, H9

MARKING GUIDELINES

Criteria	Marks
• Explains the components of scientists' methods in relation to testing and validating a model and uses specific information relating to investigations into black body radiation and photoelectric effect	4
• Explains the components of scientists' methods in relation to testing and/or validating a model refers briefly to information relating to black body radiation or photoelectric effect	3
• Describes scientists' methods with no reference to black body radiation or photoelectric effect OR • Describes black body radiation and photoelectric effect	2
• Outlines methods used by scientists OR • Describes either black body radiation or photoelectric effect	1

Question 27 (a)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Describes valid observation	2
• States procedure without describing effect	1

Question 27 (b)*Outcomes assessed: H10, H11***MARKING GUIDELINES**

Criteria	Marks
• Identifies one valid hazard and offers relevant safe work practice	2
• Identifies one valid hazard	1

Section II

Question 28 (a) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
States the role of the correction factor	1

Question 28 (a) (ii)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
- Makes a suitable inference about the geology of the survey data - Produces a table that correctly applies the three corrections to correctly reduce the gravity data for Stations B and C to produce the required gravity anomalies	4
- Makes a suitable inference about the geology of the survey data, and - Produces a data table that correctly applies two of the corrections to the gravity data but does not correctly apply the other OR - Produces a data table that correctly reduces data for either Station B or Station C	3
- Produces a data table that correctly applies the three corrections to correctly reduce the gravity data for Stations B and C to produce the required gravity anomalies OR - Makes a suitable inference about the geology of the survey data, and - Produces a data table that correctly applies at least one correction to both Stations B and C OR - Produces a data table that correctly applies at least two corrections to either Station B or C	2
- Produces a data table that correctly applies at least one correction to both Stations B and C OR - Produces a data table that correctly applies at least two corrections to either Stations B and C OR - Makes a suitable inference about the geology of the survey data	1

Question 28 (b) (i)*Outcomes assessed: H11***MARKING GUIDELINES**

Criteria	Marks
- Provides features of the procedure that increases reliability OR - Provides one feature of the procedure that increases reliability and provides a good description of the process	2
Provides one feature of the procedure that increases reliability	1

Question 28 (b) (ii)

Outcomes assessed: H8, H14

MARKING GUIDELINES

Criteria	Marks
- States that all waves follow a curved path because the density of the medium changes - Explains that S-type waves do not propagate through liquids because they are transverse waves - Explains that P-type waves do propagate through liquids because they are compression waves	4
- States that all waves follow a curved path because the density of the medium changes, and EITHER - Explains that S-type waves do not propagate through liquids because they are transverse waves OR - Explains that P-type waves do propagate through liquids because they are compression waves	3
- Explains that S-type waves do not propagate through liquids because they are transverse waves, and - Explains that P-type waves do not propagate through liquids because they are compression waves OR - States that the wave path is curved because the density of the medium changes, and - States P-type waves do propagate through liquids or states that S-type waves do not propagate through liquids	2
- States that the wave path is curved because the density of the medium changes OR - States that S-type waves do not propagate through liquids OR - States that P-type waves do propagate through liquids OR - States that outer core is liquid	1

Question 28 (c)

Outcomes assessed: H1, H2, H3, H14

MARKING GUIDELINES

Criteria	Marks
- Makes a value judgement regarding our changing understanding of Earth's geology and supports this with a detailed description of the impact of technical developments on at least two geophysical methods - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
Makes a judgement regarding our changing understanding of Earth's geology and supports this with a detailed description of the impact of technical developments on at least two geophysical methods	5–6
- Describes the geophysical methods , and - Shows the development of one method OR - Shows the impact of one method regarding our changing understanding of the Earth's geology	3–4
- Describes a geophysical method OR - Recalls the names of two geophysical methods	1–2

Question 28 (d) (i)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
Describes how an ancient imposed magnetic alignment and the Earth's current magnetic field can interact to produce either a positive (OR negative) anomaly, and indicates that the Earth's magnetic field reverses over time, and indicates that a spreading seabed will produce a pattern of magnetic anomalies	4
- Describes how an ancient imposed magnetic alignment and the Earth's current magnetic field can interact to produce either a positive (OR negative) anomaly - Indicates that the Earth's magnetic field reverses over time OR indicates how the seabed spreads to produce a pattern of magnetic anomalies	3
- Describes how an ancient imposed magnetic alignment and the Earth's current magnetic field can interact to produce either a positive or negative anomaly OR - Describes how a magnetic alignment can form in a cooling volcanic rock and indicates that the Earth's magnetic field reverses over time	2
- Identifies a magnetic anomaly as an area of different magnetic field strength OR - Identifies that igneous extruded rocks have ferromagnetic minerals OR - Identifies that the Earth's magnetic field reverses over time	1

Question 28 (d) (ii)

Outcomes assessed: H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the rate of sea-floor spreading using the specified units	3
- Identifies the position of a magnetic polarity reversal from the magnetic anomaly profile - Identifies the age of the identified magnetic polarity reversal from the magnetic polarity time scale	2
OR - Identifies the position of a magnetic polarity reversal from the magnetic anomaly profile - Identifies the age of the identified magnetic polarity reversal from the magnetic polarity time scale	1

Question 29 (a) (i)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Correctly identifies and substitutes data into appropriate equation	2
Identifies correct equation	1

Question 29 (a) (ii)

Outcomes assessed: H3, H12

MARKING GUIDELINES

Criteria	Marks
Explains reasons including supporting calculations	3
- Explains reasons with reference to only two values OR - Performs appropriate calculations without any explanation	2
- Identifies two values as giving indication of reflection OR - Explains without calculations/values	1

Question 29 (b) (i)

Outcomes assessed: H11, H12

MARKING GUIDELINES

Criteria	Marks
Provides characteristics and features of at least two appropriate criteria	3
Lists at least two appropriate criteria and provides a basic description of both	2
- Lists at least two appropriate criteria OR - Describes one appropriate criterion	1

Question 29 (b) (ii)

Outcomes assessed: H8, H9

MARKING GUIDELINES

Criteria	Marks
Describes how cancerous tissue is detected, with reference to MRI process and water content of tumour	3
- Describes MRI process OR - Outlines MRI process with reference to water content of tumour	2
- Outlines MRI process OR - States tumour has higher water content	1

Question 29 (c)

Outcomes assessed: H3, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Assesses impact of advances including clear value statement and comprehensive supporting discussion referring to CAT scan, PET scan, endoscopy and MRI - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
Comprehensive discussion of uses of EMR in medical imaging	5–6
- Assesses impact with value statement and minimal discussion (referring to 1/4 of CAT scan, PET scan, endoscopy and MRI) OR - Discussion of only a limited number of uses of EMR in medical imaging (referring to 3/4 of CAT scan, PET scan, endoscopy and MRI) without value statement	4
- Discusses impact in detail and no reference to EMR OR - Discussion of uses of EMR in medical imaging (referring to 2/4 of CAT scan, PET scan, endoscopy and MRI) without value statement	3
- Describes impact without reference to EMR OR - Describes one use of EMR in medical imaging	2
- Single value statement OR - Any correct relevant fact	1

Question 29 (d) (i)

Outcomes assessed: H3, H12

MARKING GUIDELINES

Criteria	Marks
Correctly identifies all three images	1

Question 29 (d) (ii)

Outcomes assessed: H3, H10

MARKING GUIDELINES

Criteria	Marks
Compares in detail advantages/disadvantages	3
- Identifies one advantage and one disadvantage of CAT scans over X-rays OR - Identifies two advantages OR - Identifies two disadvantages OR - Compares in detail one disadvantage/advantage	2
Identifies one advantage or one disadvantage of CAT scans over X-rays	1

Question 29 (d) (iii)

Outcomes assessed: H3, H14

MARKING GUIDELINES

Criteria	Marks
Contrasts bone scan to X-rays/CAT scans in more than one area (eg, resolution, nature of information) by describing differences or in table format	3
Compares bone scan to X-ray/CAT scan in one area by describing differences or in table format	2
States one difference between bone scan and X-ray/CAT scan	1

Question 30 (a) (i)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
• Describes observations specifically	2
• Provides a general statement referring to Doppler shift effect OR • Identifies red and blue shift	1

Question 30 (a) (ii)

Outcomes assessed: H8, H9

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates distance	3
• Identifies correct formula and substitutes appropriate values which may be in incorrect units	2
• Identifies correct formula OR • Determines period	1

Question 30 (b) (i)

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
• Provides features of the procedure that increases reliability	2
• Provides one feature of the procedure that increases reliability	1

Question 30 (b) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
- Describes an absorption spectrum and explains its production in stars involving e^- transitions - Uses the description of the spectrum to explain how the composition may be determined	4
- Describes an absorption spectrum and how it is produced, and - Identifies how a feature is used to determine composition OR - Describes an absorption spectrum, and - Gives a full description of how it is used to determine a star's composition	3
- Describes an absorption spectrum OR - Identifies how a feature is used to determine a star's composition OR - Explains how one feature is produced	2
- Identifies a feature of an absorption spectrum OR - Identifies a part of the requirements to create an absorption spectrum	1

Question 30 (c)

Outcomes assessed: H3, H10, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies several technological advances (including one or more other than adaptive/active optics or optical telescopes) - Clearly describes two or more impacts on our understanding of the cosmos arising from these advances - Makes a statement of the value of at least one technology or new understanding - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Identifies several technological advances - Clearly describes one impact on our understanding of the cosmos arising from these advances - Provides a response that demonstrates limited coherence and logical progression and includes correct use of scientific principles and ideas	4–5
- Identifies technological advances and describes the technical benefits of the identified advances OR - Identifies one or two technological advance(s) and describes the impact of one upon our understanding of the cosmos	2–3
Identifies one or more advances	1

Question 30 (d) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Correctly describes possible reactions occurring at each point	3
- Correctly describes one or two of possible reactions OR - Identifies reactions at each point	2
- Identifies one or two reactions OR - Identifies reactions as fusion reactions	1

Question 30 (d) (ii)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
Identifies that the star is on the main sequence and of approximately one solar mass, and provides suitable reason	2
- Correctly identifies type and mass of star - OR - Correctly identifies one and provides a suitable reason	1

Question 30 (d) (iii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
Identifies that a star of ten solar masses expands to a supernova and concludes as a black hole or neutron star	2
Relates this observation to the path depicted in the diagram	1

Question 31 (a) (i)

Outcomes assessed: H9, H10

MARKING GUIDELINES

Criteria	Marks
Identifies the initial state must be higher, more energetic, further out than the final state	1

Question 31 (a) (ii)

Outcomes assessed: H9, H10, H12

MARKING GUIDELINES

Criteria	Marks
Substitutes table entries into formula, correctly calculates and identifies correct means of obtaining photon energy from wavelength	4
- Substitutes table entries into formula and correctly calculates OR - Further identifies correct means of obtaining photon energy from wavelength	3
- Identifies correct formula(e) AND - Selects correct table entries	2
- Identifies correct formula OR - Selects table entries appropriate for identified transition	1

Question 31 (b) (i)

Outcomes assessed: H11, H12

MARKING GUIDELINES

Criteria	Marks
Identifies at least two sources of information	2
Outlines how reliability of information was achieved	
Outlines how reliability of information was achieved	1

Question 31 (b) (ii)

Outcomes assessed: H1, H2, H10

MARKING GUIDELINES

Criteria	Marks
- Correct statements regarding problems to be addressed by both Heisenberg and Pauli - Provides a full description of both Heisenberg's and Pauli's contribution	4
- Provides a correct statement regarding one problem to be addressed by both Heisenberg and Pauli - Provides a correct description of both Heisenberg's and Pauli's contribution	3
- Provides a correct statement regarding one problem to be addressed by either Heisenberg or Pauli, and - Provides a description of either Heisenberg's or Pauli's contribution OR - Provides a correct description of both Heisenberg's and Pauli's contribution	2
- Provides a correct statement regarding one problem to be addressed by either Heisenberg or Pauli OR - Provides an identification of either Heisenberg's or Pauli's contribution	1

Question 31 (c)

Outcomes assessed: H4, H7, H14

MARKING GUIDELINES

Criteria	Marks
- Describes in detail the fission process, including the role of the three neutrons and controlling nature of the reactor moderator - Relates production of energy to mass defect and described how this energy may be used in the production of electricity - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Describes the fission process in reactor technology and recognizes that the process in the transmutation equation results in energy production - States that thermal energy may be used in the production of electricity	3–5
- Describes the fission process OR - Identifies mass defect AND - States that thermal energy may be used in the production of electricity	1–2

Question 31 (d) (i)

Outcomes assessed: H6, H7, H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
Provides a detailed analysis of the physical principles in the operation of the cyclotron, including the effect of a magnetic field and high frequency voltage	3
Describes two physical principles in the operation of the cyclotron, stating the effect of each of these	2
- States two physical principles involved in the cyclotron OR - Describes one physical principle involved	1

Question 31 (d) (ii)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
Provides a detailed account of the use of particle accelerators in increasing our understanding of matter	2
Provides a basic account of the use of particle accelerators in increasing our understanding of matter	1

Question 31 (d) (iii)

Outcomes assessed: H10, H12

MARKING GUIDELINES

Criteria	Marks
• Identifies the composition of both the proton and neutron	2
• Identifies the composition of either the proton or the neutron	1

Question 32 (a) (i)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
• Describes structure and operation	2
• Describes structure or operation	1

Question 32 (a) (ii)

Outcomes assessed: H7, H12

MARKING GUIDELINES

Criteria	Marks
• Gives at least three valid reasons	3
• Gives two valid reasons	2
• Gives one valid reason	1

Question 32 (b) (i)

Outcomes assessed: H11, H12

MARKING GUIDELINES

Criteria	Marks
• Provides features and characteristics of at least two appropriate criteria	3
• Lists at least two appropriate criteria and provides a basic description of both	2
• Lists at least two appropriate criteria OR • States one criterion and gives a brief description	1

Question 32 (b) (ii)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
• Gives correct answer with supporting truth table	3
• Uses correct method but makes an error in one of the columns	2
• Uses correct method but makes an error in two of the columns OR • Gives correct answer without justification OR • Identifies all gates correctly	1

Question 32 (c)

Outcomes assessed: H5, H7, H14

MARKING GUIDELINES

Criteria	Marks
• Makes valid assessment • Lists three to four developments (eg, valves, transistors, ICs, storage, display technology, manufacturing techniques, packing density of components, etc) • Discusses potential future developments and the physics on which they may depend • Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5–7
• Describes three technological advances and relates them to computer performance • Makes poor reference to second part of statement	3–4
• Gives one to two advances with little or no reference to second part of statement OR • Discusses second part but not first part of the statement	2
• Mentions one development • Does not make assessment and does not refer to second part of the statement	1

Question 32 (d) (i)

Outcomes assessed: H7, H12

MARKING GUIDELINES

Criteria	Marks
Gives correct answer with justification	1

Question 32 (d) (ii)

Outcomes assessed: H7, H12

MARKING GUIDELINES

Criteria	Marks
Correctly determines sensitivity	2
- Sets up a valid ratio OR - Identifies from the graph the voltage range corresponding to the temperature range	1

Question 32 (d) (iii)

Outcomes assessed: H7, H13

MARKING GUIDELINES

Criteria	Marks
Gives correct answer, including circuit diagram	4
- Gives correct circuit diagram - Calculates required gain - Selects R_i (or R_f) but does not calculate R_f (or R_i)	3
Calculates gain and appropriate resistors for an incorrect circuit	2
- Identifies gain formula OR - Identifies feedback circuit element	1